
Stochastic Actor-Critic: Mitigating Overestimation Bias with a Single Distributional Critic

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Abstract

Off-policy actor-critic methods in reinforcement learning train a critic with temporal-difference updates and use it as a learning signal for the policy (actor). This design typically achieves higher sample efficiency than purely on-policy methods. However, critic networks tend to overestimate value estimates systematically. This is often addressed by introducing a pessimistic bias based on uncertainty estimates. Current methods employ ensembling to quantify the critic’s epistemic uncertainty (arising from limited data) to scale pessimistic updates. In this work, we propose a new algorithm called Stochastic Actor-Critic (STAC) that incorporates temporal (one-step) aleatoric uncertainty (due to noise in the data) of the critic to scale pessimistic bias in the temporal-difference targets, rather than relying on epistemic uncertainty. STAC uses a single distributional critic network to model the temporal distribution of return, and applies dropout to both the critic and actor networks for regularization. Our results show that pessimism based on a distributional critic alone suffices to mitigate overestimation, and leads to risk-averse behavior in stochastic environments. Introducing dropout further improves training stability and performance by means of regularization. With this combination, STAC achieves better computational efficiency using a single distributional critic network.

1 Introduction

Actor-critic methods leverage off-policy samples to train critics, promising higher sample-efficient learning than on-policy algorithms. Despite this advantage, actor-critic agents often struggle with *overestimation bias* in value function learning due to the joint effects of function approximation, temporal-difference learning, and off-policy sampling (Thrun & Schwartz, 2014; Sutton & Barto, 2018; Van Hasselt et al., 2018), which can destabilize training. In this work, we revisit this problem from a fresh perspective, focusing on the use of uncertainty modeling in actor-critic architectures.

Epistemic uncertainty indicates a lack of training data and is used by current actor-critic methods to scale pessimistic updates (Fujimoto et al., 2018; Haarnoja et al., 2018; Kuznetsov et al., 2020; Moskovitz et al., 2021; Chen et al., 2021; Hiraoka et al., 2021; Zhang et al., 2024). These methods use critic ensembles for this kind of uncertainty modeling. However, pessimistic updates based on epistemic uncertainty in the critic may hinder exploration of the state-action space, contradicting the principle of *optimism in the face of uncertainty* (Kocsis & Szepesvári, 2006; Audibert et al., 2007; Azizzadenesheli et al., 2018; Ciosek et al., 2019; O’Donoghue, 2023; Wu et al., 2023).

On the other hand, aleatoric uncertainty is a measure of noise within training data. Some works propose to use pessimistic updates based on both epistemic and aleatoric uncertainty of return (discounted cumulative reward) for overestimation mitigation (Kuznetsov et al., 2020). Originally, modeling aleatoric uncertainty is a topic of distributional reinforcement learning (Bellemare et al., 2017; Dabney et al., 2018b; Kim & Oh, 2022; Duan et al., 2021; 2025; Ma et al., 2025), and mainly used to scale risk sensitivity of the agent (Tang et al., 2019; Yang et al., 2021; Théate & Ernst, 2023; Ma et al., 2025). While pessimistic policy updates lead to risk-averse behavior, optimistic updates result in risk-seeking behavior. Recently, in Q-learning setting,

Achab et al. (2023) proposed to model temporal aleatoric distribution of return, in which uncertainty is induced by only one-step dynamics of the environment, rather than across all steps. This approach is proven to have convergence guarantees theoretically.

In off-policy actor-critic methods, the temporal-difference target changes continuously as the policy is updated. Although policy improvement is deterministic given a critic, high-dimensional function approximation causes small critic errors to be selectively exploited in a non-stationary manner, inducing irreducible variability in value targets even in deterministic environments, which is a major source of overestimation. We argue that this variability is best captured as temporal aleatoric uncertainty in the critic output, as it aggregates all uncertainty sources contributing to overestimation: policy-induced effects and transition or reward stochasticity.

In this work, we show that temporal aleatoric uncertainty can be modeled by a distributional critic, and that pessimism applied to this uncertainty alone is sufficient to control overestimation—without ensembles or multiple critics. Based on this idea, we introduce a novel off-policy distributional actor-critic algorithm, Stochastic Actor-Critic (STAC), specifically modeling one-step return uncertainty for critic, and pessimistic updates based on this distribution. This way, STAC eliminates the need for double/ensemble critics, reducing both computation and memory costs. In other words, STAC uses *pessimism in the face of temporal aleatoric uncertainty* for overestimation mitigation.

Our method differs from other methods using aleatoric uncertainty for risk aversion Tang et al. (2019); Yang et al. (2021); Kim & Oh (2022); Ma et al. (2025). These methods, incorporate pessimism on total return uncertainty only for actor updates, whereas STAC uses pessimism on temporal uncertainty for both critic and actor updates. The main purpose is overestimation mitigation rather than risk-averse learning, but pessimism in STAC still yields to risk aversion, similar to fully distributional methods.

Recent work of Nauman et al. (2024) has also shown that network regularization has positive impact on overestimation by reducing over-fitting. Ensemble based methods inherently regularized by using multiple critic networks. Since STAC uses a single critic network, it is sensitive to learning errors and over-fitting. For this purpose, dropout (Srivastava et al., 2014) and layer normalization (Ba et al., 2016) are employed in both actor and critic networks as regularization tools.

The implementation is very simple and can be obtained by introducing dropout to networks, modeling a distributional critic and defining a pessimistic learning objective upon Soft Actor-Critic algorithm (Haarnoja et al., 2018). We conduct experiments on standard RL benchmarks to evaluate the performance of STAC compared to existing methods. Our results demonstrate the effectiveness of STAC in achieving competitive performance with state-of-the-art methods while requiring fewer computational resources (single critic), making it a promising approach for real-world RL applications.

2 Preliminaries

This section introduces the minimum background required to analyze uncertainty-induced overestimation in off-policy actor-critic methods. Throughout the paper, $\mathcal{P}(\Omega)$ denotes the set of all possible probability distributions on the set Ω . $A \stackrel{D}{=} B$ indicates that two random variables, A and B , have identical probability laws.

2.1 Aleatoric and Epistemic Uncertainty

Uncertainty in an estimate is mainly categorized as either *aleatoric* or *epistemic* (Der Kiureghian & Ditlevsen, 2009; Kendall & Gal, 2017; Gal et al., 2016b). Epistemic uncertainty refers to uncertainty of model parameters (θ) due to insufficient training data. Given the training data $\mathcal{D}$ and prior distribution $p(\theta)$ over the network parameters θ , we can compute the posterior distribution over the parameters $p(\theta | \mathcal{D})$ using Bayesian inference.

On the other hand, aleatoric uncertainty is induced by the inherent randomness within the data. Even with infinite data, it is unavoidable and cannot be reduced, because it is an intrinsic part of the process being modeled. In deep learning, we can model this by having a distributional network that outputs a probability distribution $p_\theta(y|x)$ conditioned on input x (Lakshminarayanan et al., 2017; Kendall & Gal, 2017). Given

a dataset $\mathcal{D}$, the loss function for training the network can be derived from the negative log-likelihood: $\mathcal{L}_\theta(\mathcal{D}) = \mathbb{E}_{\{(x_i, y_i)\} \sim \mathcal{D}} [-\log p_\theta(y_i | x_i)]$. In reinforcement learning, aleatoric uncertainty naturally arises from stochastic rewards, transition dynamics, and policy-induced randomness in temporal-difference targets.

2.2 Maximum Entropy Actor-Critic Reinforcement Learning

In reinforcement learning language, the agent lives in a Markov Decision Process (MDP) which is represented by a tuple $\mathcal{M} = (\mathcal{S}, \mathcal{A}, d_0, \tau, R)$, where $\mathcal{S}$ is the state space, $\mathcal{A}$ is the action space, $d_0 \in \mathcal{P}(\mathcal{S})$ is the initial state distribution, $\tau : \mathcal{S} \times \mathcal{A} \rightarrow \mathcal{P}(\mathcal{S})$ is the transition kernel and $R : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ is the reward function.

The initial state is sampled first, $s_0 \sim d_0(\cdot)$. At time step t , being in state s_t ; next state is obtained from the environment, $s_{t+1} \sim \tau(\cdot | s_t, a_t)$ depending on the taken action $a_t \sim \pi(\cdot | s_t)$. Finally, a reward is obtained, $r_t = R(s_t, a_t)$ from the reward function R . Return and action return are defined respectively as follows,

$$G^\pi(s) \stackrel{\text{D}}{=} \sum_{t=0}^{\infty} \gamma^t (R(s_t, a_t) - \alpha \log \pi(a_t | s_t)), \quad s_0 = s, \quad (1)$$

$$Z^\pi(s, a) \stackrel{\text{D}}{=} R(s, a) + \sum_{t=1}^{\infty} \gamma^t (R(s_t, a_t) - \alpha \log \pi(a_t | s_t)), \quad s_0 = s, \quad a_0 = a, \quad (2)$$

The ultimate goal of the agent is to derive a policy $\pi : \mathcal{S} \rightarrow \mathcal{P}(\mathcal{A})$ to maximize discounted return G^π with entropy bonus (Haarnoja et al., 2017; 2018). However, returns are random variables, and cannot be used as objective. Therefore, value (V) and action-value (Q) functions are defined as expectations of return over policy and transition dynamics, $V^\pi(s) = \mathbb{E}_{\pi, \tau} [G^\pi(s)]$, $Q^\pi(s, a) = \mathbb{E}_{\pi, \tau} [Z^\pi(s, a)]$. Actor-critic methods model action value function Q , and learning iterates between solving policy evaluation and policy improvement.

Policy evaluation minimizes the temporal difference: $\|\mathcal{T}^\pi Q(s, a) - Q(s, a)\|$ by a gradient step, where $\mathcal{T}^\pi Q$ is the Bellman backup operator, and defined as,

$$\mathcal{T}^\pi Q(s, a) = R(s, a) + \gamma \mathbb{E}_{\substack{s' \sim \tau(\cdot | s, a) \\ a' \sim \pi(\cdot | s')}} [Q(s', a') - \alpha \log \pi(a' | s')]. \quad (3)$$

Policy improvement is minimizing value function $V^\pi(s) = \mathbb{E}_{a \sim \pi} [Q^\pi(s, a) - \alpha \log \pi(a | s)]$ by a gradient step. After policy improvement and policy evaluation at step k , updated action value function at step $k+1$ is $Q^{k+1}(s, a) = \mathcal{T}^* Q^k(s, a)$, where $\mathcal{T}^*$ is the Bellman optimality operator (Equation 5 from Haarnoja et al. (2017)),

$$\mathcal{T}^* Q(s, a) = R(s, a) + \gamma \mathbb{E}_{s' \sim \tau(\cdot | s, a)} \left[\tilde{\alpha} \log \left(\int_{\mathcal{A}} \exp(\tilde{\alpha}^{-1} Q(s', a')) da' \right) \right]. \quad (4)$$

$\tilde{\alpha} \in [\alpha, \infty)$ is defined to demonstrate the continuum of policy improvement, i.e., the fact that policy is updated by partially exploiting state-action value function at each iteration. As $\tilde{\alpha} \rightarrow \infty$, $\mathcal{T}^* Q$ boils down to $\mathcal{T}^u Q$, which represents action value for uniform policy. On the other hand, $\tilde{\alpha} = \alpha$ represents updated action value after complete exploitation. Consequently, in deep learning setting, a higher learning rate of actor network results in lower $\tilde{\alpha}$, closer to α .

3 Distributional Reinforcement Learning and Overestimation

3.1 Distributional Maximum Entropy Actor-Critic

Instead of learning action value function Q , action return can also be modeled as a random variable to identify possible consequences of a given policy (Bellemare et al., 2017). This also constitutes the methodology of distributional maximum entropy actor-critic methods (Duan et al., 2021; Ma et al., 2025). For this,

action return distribution $\mathcal{Z} \in \mathcal{P}(\mathbb{R}^{S \times \mathcal{A}})$ is defined, where action returns are sampled from this distribution, $Z(s, a) \sim \mathcal{Z}(s, a)$. Recall that $Q(s, a) = \mathbb{E}_{\mathcal{Z}, \tau}[Z(s, a)]$. The corresponding Bellman backup operator $\mathcal{T}^\pi$ is defined as,

$$\mathcal{T}^\pi Z(s, a) \stackrel{\text{D}}{=} R(s, a) + \gamma(Z(s', a') - \alpha \log \pi(a' | s')), \quad s' \sim \tau(\cdot | s, a), \quad a' \sim \pi(\cdot | s'), \quad (5)$$

To make notation easier, distributional Bellman backup $\mathcal{T}_D^\pi \mathcal{Z}$ is defined, where Bellman backup of sampled values are sampled from this distribution, i.e., $\mathcal{T}^\pi Z(s, a) \sim \mathcal{T}_D^\pi \mathcal{Z}(s, a)$. At each step k , the state-action return distribution is updated by minimizing Kullback-Leibler divergence from distributional Bellman backup: $D_{\text{KL}}(\mathcal{T}_D^\pi \mathcal{Z}(s, a) || \mathcal{Z}(s, a))$ by a gradient step.

One-step Distributional Uncertainty Unlike conventional distributional modeling, it is also possible to model action return with randomness only up to first step Achab (2020); Achab et al. (2023). In this approach, Bellman backup uses expectation (deterministic) of return of next state and action. This way, uncertainty due to environment stochasticity and actor-induced uncertainty (due to policy improvement) are modeled for only one step, which directly targets the source of critic overestimation. This distribution is denoted as $\mathcal{Q}$ instead of $\mathcal{Z}$. Therefore, $Q \sim \mathcal{Q}$ is a random variable unlike previous formulation. Bellman backup is defined similar as in Equation 5 but expectation of next-state value is used,

$$\mathcal{T}^\pi Q(s, a) \stackrel{\text{D}}{=} R(s, a) + \gamma(\mathbb{E}_{Q \sim \mathcal{Q}}[Q(s', a')] - \alpha \log \pi(a' | s')), \quad s' \sim \tau(\cdot | s, a), \quad a' \sim \pi(\cdot | s'). \quad (6)$$

At each step k , the state-action value distribution is updated by minimizing Kullback-Leibler divergence from distributional Bellman backup: $D_{\text{KL}}(\mathcal{T}_D^\pi Q(s, a) || Q(s, a))$ by a gradient step. Corresponding Bellman optimality operator is as follows,

$$\mathcal{T}^* Q(s, a) \stackrel{\text{D}}{=} R(s, a) + \gamma \mathbb{E}_{Q \sim \mathcal{Q}} \left[\tilde{\alpha} \log \left(\int_{\mathcal{A}} \exp(\tilde{\alpha}^{-1} Q(s', a')) da' \right) \right], \quad s' \sim \tau(\cdot | s, a). \quad (7)$$

This operator is proved to be a contraction in Q-learning by Achab et al. (2023), using max instead of logsumexp operator. This guarantees learning convergence in tabular setting. However, in function approximation, there are other phenomena like overestimation and over-fitting. As a main focus of this work, we will discuss overestimation bounds of one-step Bellman optimality operator, and propose a method to mitigate it in the next section.

3.2 Quantifying Overestimation for Sub-Gaussian Critic Distributions

Let $\mu(s, a) = \mathbb{E}_{\mathcal{Q}, \tau}[Q(s, a)]$, the Bellman optimality backup of the mean is not equal to the mean of the backup, $\mathcal{T}^* \mu(s, a) \neq \mathbb{E}_{\mathcal{Q}, \tau}[\mathcal{T}^* Q(s, a)]$ due to policy improvement, and the difference is the overestimation bias. In this section, we analyze the overestimation bias similar to Chen et al. (2021) and Lan et al. (2020) but in the soft learning framework instead of discrete actions. We define the overestimation error as the difference between $\mathbb{E}_{\mathcal{Q}, \tau}[\mathcal{T}^* Q(s, a)]$ and average $\mathcal{T}^* \mu(s, a)$ as ϵ ,

$$\epsilon(s, a) = \mathbb{E}_{\mathcal{Q}, \tau}[\mathcal{T}^* Q(s, a)] - \mathcal{T}^* \mu(s, a). \quad (8)$$

In the ideal case, $\epsilon(s, a)$ should be zero if there is no overestimation, which is not the case due to critic uncertainty. To quantify it, we assume that critic distribution $\mathcal{Q}(s, a)$ is sub-Gaussian with variance proxy $\sigma^2(s, a)$ and mean $\mu(s, a)$. We adopt a sub-Gaussian assumption as it provides a mild and widely used concentration model that enables analytic overestimation bounds without restricting the critic to a specific parametric distribution.

Definition 3.1. A random variable $X \in \mathbb{R}$ with mean $\mu = \mathbb{E}[X]$ is called sub-Gaussian with variance proxy σ^2 if its moment generating function satisfies

$$\mathbb{E}[\exp(\lambda X)] \leq \exp \left(\lambda \mu + \frac{1}{2} \lambda^2 \sigma^2 \right), \quad \forall \lambda \in \mathbb{R}. \quad (9)$$

Under the sub-Gaussian assumption, Theorem 3.1 quantifies how critic uncertainty propagates through the soft Bellman optimality operator and Corollary 3.1.1 motivates a variance-dependent pessimistic correction.

Theorem 3.1 (Overestimation quantification for sub-Gaussian critics). *Let given state-action value distribution $\mathcal{Q} \in \mathcal{P}(\mathbb{R}^{S \times A})$ be sub-Gaussian with mean $\mu(s, a)$ and variance proxy $\sigma^2(s, a)$ for all state-action pairs, with bounded support. For each sample $Q \sim \mathcal{Q}$,*

$$\mathcal{T}^*Q(s, a) \leq R(s, a) + \gamma \tilde{\alpha} \log \left(\int_A \exp(\tilde{\alpha}^{-1} \mu(s', a') + \frac{1}{2} \tilde{\alpha}^{-2} \sigma^2(s', a')) da' \right). \quad (10)$$

Corollary 3.1.1 (Pessimistic critic target). *For policy evaluation, using pessimistic Bellman backup,*

$$\mathcal{T}_\beta^\pi Q(s, a) \stackrel{D}{=} R(s, a) + \gamma (\mu(s', a') - \beta \sigma(s', a') - \alpha \log \pi(a' | s')), \quad s' \sim \tau(\cdot | s, a), \quad a' \sim \pi(\cdot | s'), \quad (11)$$

prevents overestimation as long as $\beta \geq \max_{(s, a)} \frac{1}{2} \tilde{\alpha}^{-1} \sigma(s, a)$.

Moreover, maximum overestimation bound is demonstrated in Theorem 3.2.

Theorem 3.2 (Overestimation bound). *Overestimation due to uncertainty of distribution $\mathcal{Q}$, denoted as ϵ , is upper bounded for Bellman updates,*

$$\epsilon(s, a) \leq \frac{\gamma}{2\tilde{\alpha}} \mathbb{E}_{s' \sim \tau} \left[\max_{a'} \sigma^2(s', a') \right]. \quad (12)$$

As an additional notation, we introduce pessimistic distributional Bellman backup $\mathcal{T}_{\beta, D}^\pi \mathcal{Q}$ where pessimistic Bellman backup of sampled values are sampled from this distribution, i.e., $\mathcal{T}_\beta^\pi Q(s, a) \sim \mathcal{T}_{\beta, D}^\pi \mathcal{Q}(s, a)$. The difference between $\mathcal{T}_{\beta, D}^\pi \mathcal{Q}(s, a)$ and $\mathcal{T}_D^\pi \mathcal{Q}(s, a)$ is illustrated in Figure 1.

Taken together, these results show that overestimation in off-policy actor-critic methods originates from temporal uncertainty in Bellman targets, and that a simple variance-proxy-based pessimistic shift is sufficient to control it. This observation motivates the Stochastic Actor-Critic algorithm introduced next.

4 Stochastic Actor-Critic

In this section, we discuss key mechanisms needed for computation and sample efficient actor-critic learning and propose our algorithm *Stochastic Actor-Critic*. This algorithm employs a *single distributional* critic network $\mathcal{Q}_\theta$ by parameter set θ which captures temporal (one-step) aleatoric uncertainty, and dropout along with layer normalization for network regularization. Actor network π_ϕ outputs a **tanh** transformed normal distribution to bound actions to $[-1, 1]$, and is parameterized by set ϕ . Unlike other methods, STAC uses critic prediction in a pessimistic manner using temporal aleatoric uncertainty, for policy evaluation and policy improvement.

The rationale behind this argument is that most of the temporal difference target randomness due to environment stochasticity and ongoing actor updates (policy improvement) inherently appears in the form of aleatoric uncertainty from the critic’s side, and overestimation is the result of this randomness. Although policy improvement itself is deterministic given a critic, small approximation errors in the critic are selectively exploited by the policy update in a non-stationary manner, which induces unpredictable variability in Bellman targets in practice. Unlike distributional methods (Bellemare et al., 2017), we do not model aleatoric uncertainty of the full return, since it is not directly related to overestimation.

Although return uncertainty is modeled only for one-step, pessimism in STAC also leads to risk-averse learning since critic is also learned by pessimistic updates. Conventional distributional algorithms model full return as a distribution. For risk-averse learning only actor is updated in pessimistic way, not critic, usually by conditional value at risk measure (CVaR) (Dabney et al., 2018a; Tang et al., 2019; Yang et al., 2021; Kim & Oh, 2022; Ma et al., 2025) or by standard deviation (Ma et al., 2025).

Epistemic pessimism naturally decreases the degree of overestimation by decreasing probability of critic errors which are exploited by policy improvement later. However, out-of-distribution states and actions are

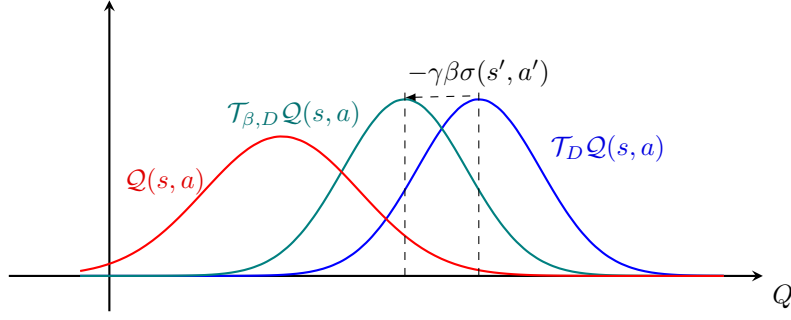


Figure 1: Evolution of the Pessimistic Distributional Bellman Backup. While $\mathcal{T}_D^\pi Q(s, a)$ is overestimated, a corrected backup $\mathcal{T}_{\beta, D}^\pi Q(s, a)$ is closer to $Q(s, a)$.

detected by high epistemic uncertainty, and should not be suppressed by pessimism. On the contrary, these state-action pairs should be visited for exploration. Therefore, epistemic uncertainty should not be used as a pessimism signal for overestimation mitigation unlike previous methods.

(Nauman et al., 2024) demonstrated that network regularization methods improve performance. Since STAC employs a single critic network, it is prone to over-fitting more than ensemble based methods. To prevent it, STAC employs dropout (Srivastava et al., 2014) similar to Hiraoka et al. (2021). While dropout may also promote exploration by injecting stochasticity into the policy and value estimates, we do not attempt to isolate or prove this effect here. Dropout also allows capturing the probabilistic nature of a network, representing Bayesian neural networks (Gal & Ghahramani, 2016). However, it is important to note that STAC employs dropout on both critic and actor networks only for regularization purposes, not for epistemic modeling.

4.1 Policy Evaluation

For simplicity, STAC models critic as normal distribution. This contradicts the bounded distribution assumption of Theorem 3.1, but it yields a simple loss function and is easy to interpret. Still, it is reasonable to assume critic distribution is bounded for finite horizon or discounted MDPs ($\gamma < 1$) with bounded reward functions.

Using transition tuples from experience replay as batch, $\mathcal{D}_b = \{(s_i, a_i, r_i, s'_i, \text{done}_i)\}_{i=1}^{N_b}$, temporal difference (TD) target Q_i^{TD} , representing Bellman backup, is β -pessimistic,

$$Q_i^{TD} = r_i + \gamma(\mu_{\bar{\theta}}(s'_i, \tilde{a}'_i) - \beta\sigma_{\bar{\theta}}(s'_i, \tilde{a}'_i) - \alpha \log \pi_{\phi}(s'_i, \tilde{a}'_i))(\neg \text{done}_i), \quad \tilde{a}'_i \sim \pi_{\phi}(\cdot | s'_i). \quad (13)$$

Learning objective is cross-entropy loss (log loss),

$$\mathcal{L}_{\theta}(\mathcal{D}_b) = \frac{1}{N_b} \sum_{i=1}^{N_b} -\log \mathcal{Q}_{\theta}(Q_i^{TD} | s_i, a_i) = \frac{1}{2} \log 2\pi + \frac{1}{2N_b} \sum_{i=1}^{N_b} \left(\log \sigma_{\theta}^2(s_i, a_i) + \frac{(Q_i^{TD} - \mu_{\theta}(s_i, a_i))^2}{\sigma_{\theta}^2(s_i, a_i)} \right). \quad (14)$$

4.2 Policy Improvement

According to Corollary 3.1.1, pessimistic TD targets should be used to train critic network. However, this analysis does not account for policy learning since policy is assumed as softmax over action values. In actor-critic framework, same pessimistic objective should be used for policy improvement, since policy evaluation and policy improvement objective should be the same. In other words, at learning step k , Bellman backup must be equal to Bellman policy evaluation with the new policy, $\mathcal{T}^* Q^k \stackrel{D}{=} \mathcal{T}^{\pi^{k+1}} Q^k$. It is only possible by using same pessimistic critic value for policy improvement.

4.3 Other Details and Algorithm Summary

Layer Normalization We implement Layer Normalization (Ba et al., 2016) after all hidden activations of critic and actor networks, in order to stabilize learning and prevent possible numerical instabilities.

Lagged critic for TD target When the trained critic network is also used in calculating the target value, the critic training is prone to divergence (Li et al., 2023). For this, a common approach is to use another critic network to evaluate TD target (Mnih et al., 2013). Similar to Lillicrap et al. (2015), Fujimoto et al. (2018), and Haarnoja et al. (2018), we use a delayed form of critic network for TD target evaluations as demonstrated in Equation 13. The parameters of target critic are only updated by Polyak averaging of main critic network weights through learning steps; $\bar{\theta} \leftarrow \rho\bar{\theta} + (1 - \rho)\theta$. This strategy is important to ensure the stability of temporal difference learning.

Automatic temperature tuning Using constant temperature results in different policies if the reward magnitude changes. To mitigate this, Haarnoja et al. (2018) proposed a policy entropy constraint, representing temperature as the Lagrange multiplier of the constraint. Given target entropy $\bar{\mathcal{H}}$ as hyper-parameter, the loss function related to this constraint is as follows;

$$\mathcal{L}_\alpha(\mathcal{D}_b) = -\alpha\bar{\mathcal{H}} + \alpha \sum_{i=1}^{N_b} \mathbb{E}_{a \sim \pi_\phi(\cdot | s_i)} [-\log \pi_\phi(a | s_i)]. \quad (15)$$

Networks illustrations are available in Appendix D. Note that the bar notation stands for the lagged network with non-trainable parameters. STAC is summarized in Algorithm 1 with gradient descent but Adam optimizer (Kingma & Ba, 2014) is used in our experiments.

Algorithm 1 Stochastic Actor-Critic

Require: Environment `env`

Require: Experience buffer $\mathcal{D}$

Require: Critic $\mathcal{Q}_\theta$, lagged critic $\mathcal{Q}_{\bar{\theta}}$, actor π_ϕ , all with dropout

Require: Initial temperature α , target entropy $\bar{\mathcal{H}}$

Require: Pessimism β

Require: Learning rates $\eta_Q, \eta_\pi, \eta_\alpha$, Polyak parameter ρ

Require: Total training steps N , batch size N_b

```

 $s \sim \text{env.reset}()$  ▷ Reset the environment
for  $N$  timesteps do
   $a \sim \pi_\phi(\cdot | s)$  ▷ Sample action
   $r, s', \text{done} \sim \text{env.step}(a)$  ▷ Act on environment
   $\mathcal{D} \leftarrow \mathcal{D} \cup (s, a, r, s', \text{done})$  ▷ Record transition tuple
  if  $\text{done}$  then  $s \leftarrow s'$  else  $s \sim \text{env.reset}()$  ▷ State transition or reset
  for  $G$  gradient steps do
     $\mathcal{D}_b = \{(s_i, a_i, r_i, s'_i, \text{done}_i)\}_{i=1}^{N_b} \sim \mathcal{D}$  ▷ Sample minibatch for training
     $\tilde{a}'_i \sim \pi_\phi(\cdot | s'_i)$  ▷ Sample next actions
     $Q_i^{TD} = r_i + \gamma(\mu_{\bar{\theta}}(s'_i, \tilde{a}'_i) - \beta\sigma_{\bar{\theta}}(s'_i, \tilde{a}'_i) - \alpha \log \pi_\phi(\tilde{a}'_i | s'_i))(\neg \text{done}_i)$  ▷ Build TD targets
     $\theta \leftarrow \theta - \eta_Q \nabla_\theta \left( \frac{1}{N_b} \sum_{i=1}^{N_b} -\log \mathcal{Q}_\theta(Q_i^{TD} | s_i, a_i) \right)$  ▷ Update critic
     $\phi \leftarrow \phi - \eta_\pi \nabla_\phi \left( \frac{1}{N_b} \sum_{i=1}^{N_b} \mathbb{E}_{a \sim \pi_\phi(\cdot | s_i)} [\mu_\theta(s_i, a) - \beta\sigma_\theta(s_i, a) - \alpha \log \pi_\phi(a | s_i)] \right)$  ▷ Update actor
     $\alpha \leftarrow \alpha - \eta_\alpha \nabla_\alpha \left( -\alpha\bar{\mathcal{H}} + \alpha \sum_{i=1}^{N_b} \mathbb{E}_{a \sim \pi_\phi(\cdot | s_i)} [-\log \pi_\phi(a | s_i)] \right)$  ▷ Update temperature
     $\bar{\theta} \leftarrow \rho\bar{\theta} + (1 - \rho)\theta$  ▷ Update target critic network
  end for
end for

```

5 Experiments

Through Gymnasium API (Towers et al., 2023), six MuJoCo and three Box2D environments are used for experiments as they are tested by most algorithms in the literature. Box2D environments are inherently random, **BipedalWalker-v3** and **BipedalWalkerHardcore-v3** models a two-legged robot on randomly generated terrain, while **LunarLander-v3** models a landing spaceship under wind and turbulence, which is maximized in our experiments.

For evaluation, after each 1000 time steps, we execute a single test episode using the online policy and measure its performance by calculating the total reward accumulated during the episode. Specified environments are trained through a fixed number of environment interactions, repeated 5 times to assess the stability of the algorithm. Further experimental details are presented in Appendix C.

Hyper-parameters per environment can be found in Table 4 of Appendix C. For all experiments, PyTorch (version 2.7.1) (Paszke et al., 2019) is used. Please refer to Appendix E for the codebase.

5.1 Comparison to Other Algorithms

Our experiments aim to investigate whether enhancing off-policy actor-critic methodology with STAC can improve their sample and computation efficiency on continuous-control benchmarks. For this purpose, STAC is compared to similar competitive algorithms; Distributional Soft-Actor Critic (DSAC) (Ma et al., 2025), and Soft Actor-Critic (SAC) (Haarnoja et al., 2018). Both methods use minimum of double critics for updates. All algorithm results are obtained using in-house code with the same network architectures (including layer normalization but not dropout) to make a fair comparison.

We also add another algorithm called Epistemic Stochastic Actor-Critic (ESTAC), with double distributional critic with one-step uncertainty similar to STAC. For overestimation, minimum of two critic is used for learning similar to SAC and DSAC, instead of temporal aleatoric pessimism. The purpose here is to discover source of improvements of STAC over other methods, and the only difference between ESTAC and STAC is overestimation mitigation mechanism.

The pessimism rates and dropout configurations used for comparison are resulted from ablation studies, and explained in the next section. Optimal configurations are summarized in Table 5.

Learning curves In Figure 2, the performance of STAC is shown against previously mentioned state-of-the-art algorithms. Additionally, value estimation errors are presented in Figure 3. The bold lines represent the inter-quartile mean, while the shaded area indicates the quartiles (to represent randomness through seeds) of the total reward across evaluation episodes. Average returns through all learning processes averaged over random seeds are summarized in Table 1 and final performance is summarized in Table ??.

Table 1: Average episodic return through learning procedure and over five training runs on MuJoCo and Box2D tasks. $\pm$ sign represents mean and standard deviation of overall score.

Env	# steps	DSAC	ESTAC	SAC	STAC
Ant-v4	3M	5402.8 $\pm$ 2427.5	4369.8 $\pm$ 2289.2	5676.6$\pm$2032.5	5088.0 $\pm$ 2295.8
BipedalWalker-v3	1M	293.6$\pm$115.1	262.7 $\pm$ 123.1	245.7 $\pm$ 150.2	261.5 $\pm$ 113.3
BipedalWalkerHardcore-v3	3M	51.7 $\pm$ 26.4	37.7 $\pm$ 100.7	-41.7 $\pm$ 83.3	83.5$\pm$143.2
HalfCheetah-v4	3M	10944.0 $\pm$ 3049.7	10320.4 $\pm$ 2640.7	9363.6 $\pm$ 2136.8	11291.8$\pm$3095.7
Hopper-v4	1M	1629.1 $\pm$ 1022.4	1354.0 $\pm$ 1087.9	1998.2 $\pm$ 1238.5	2614.3$\pm$1167.4
Humanoid-v4	3M	6425.3$\pm$2707.3	6034.8 $\pm$ 2738.5	5179.4 $\pm$ 2161.3	6105.4 $\pm$ 2739.0
LunarLander-v3	1M	199.5 $\pm$ 140.8	154.2 $\pm$ 166.0	212.6$\pm$138.2	201.6 $\pm$ 135.0
Swimmer-v4	1M	57.7 $\pm$ 27.0	76.7 $\pm$ 26.2	57.3 $\pm$ 23.2	97.7$\pm$32.9
Walker2d-v4	1M	4867.9 $\pm$ 1582.5	4934.8$\pm$1596.1	3935.1 $\pm$ 1571.9	4829.7 $\pm$ 2383.8

Sample efficiency As seen from Figure 2, Table 1 and Table 2, STAC outperforms other algorithms on some environments in terms of sample efficiency (see **BipedalWalkerHardcore-v3**, **HalfCheetah-v4**, **Hopper-v4**, **Swimmer-v4**). However, it is important to note that there are many factors that affects performance, such

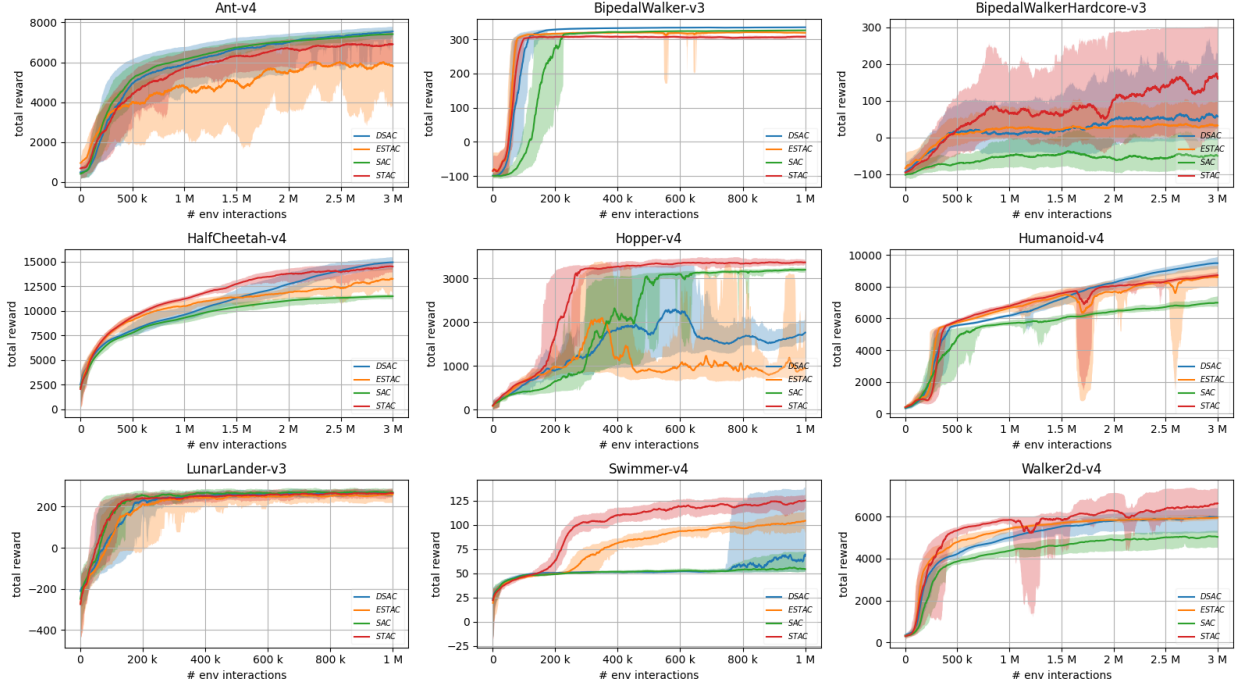


Figure 2: Episodic score curves of STAC and other algorithms.

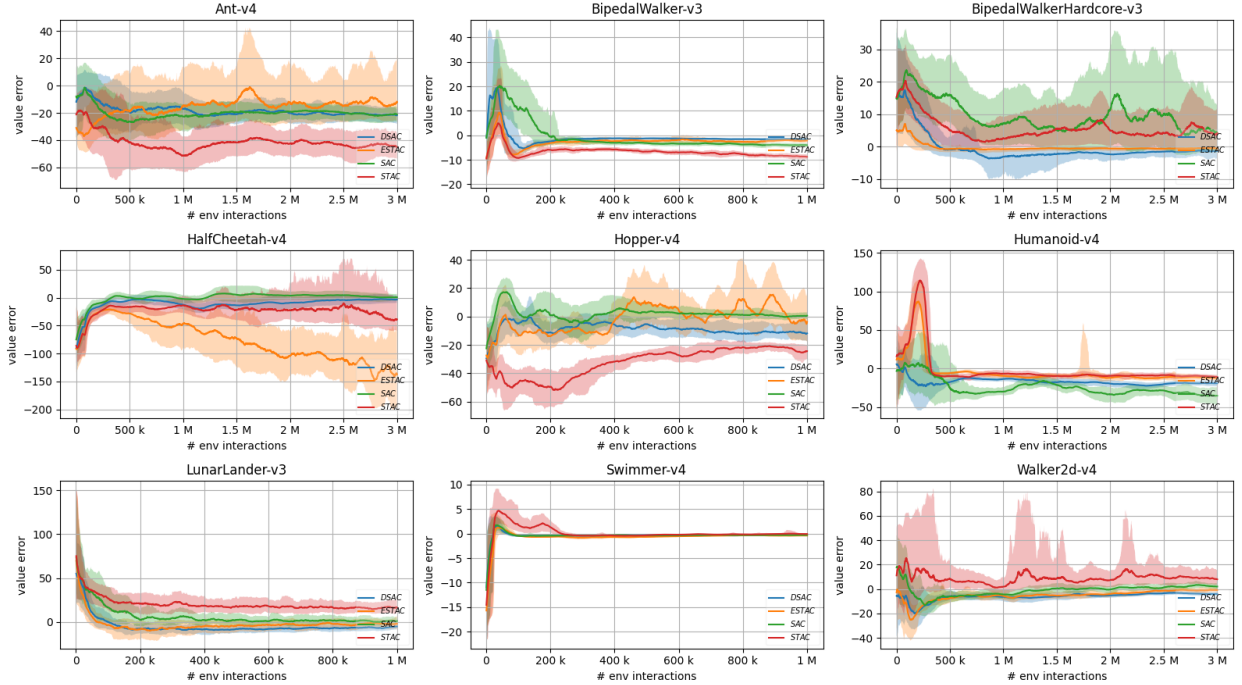


Figure 3: Average episodic value estimation error of STAC and other algorithms.

Table 2: Inter-quartile mean results of last %1 of evaluation episodes, β and dropout for STAC is selected with best performance.

Env	# steps	DSAC	ESTAC	SAC	STAC
Ant-v4	3M	7543.1	5798.9	7407.1	6907.3
BipedalWalker-v3	1M	335.3	319.2	326.3	308.0
BipedalWalkerHardcore-v3	3M	56.8	32.3	-50.8	159.9
HalfCheetah-v4	3M	14921.5	13241.6	11475.6	14489.5
Hopper-v4	1M	1762.9	944.3	3196.4	3363.7
Humanoid-v4	3M	9484.1	8608.9	6990.4	8726.2
LunarLander-v3	1M	265.4	258.3	268.9	265.7
Swimmer-v4	1M	69.3	104.1	54.2	125.0
Walker2d-v4	3M	5981.4	5952.5	5036.4	6634.7

as distribution modeling method, number of critics and regularization method, network size and learning rates etc. Therefore, rather than performance, we demonstrate that STAC has a competitive performance, no convergence issue, and achieves this by using single critic while others use double.

5.2 Sweeping Pessimism and Dropout

STAC is experimented by varying levels of pessimism β and dropout rates to isolate effect of pessimism and dropout individually. We used 5 pessimism level and 4 dropout configurations: no dropout, actor dropout, critic dropout and actor&critic dropout.

Table 3: Inter-quartile mean scores of last %1 of evaluation episodes, for varying β . Dropout rate is 0.01 for critic and actor.

Env	$\beta = 0.0$	$\beta = 0.125$	$\beta = 0.25$	$\beta = 0.375$	$\beta = 0.5$
Ant-v4	5264.4	7082.7	6907.3	6246.4	4834.4
BipedalWalker-v3	171.9	125.4	262.6	308.0	305.1
BipedalWalkerHardcore-v3	61.8	118.8	43.4	-0.0	-59.8
HalfCheetah-v4	14489.5	12299.6	13385.4	11042.1	7602.7
Hopper-v4	1132.4	1308.8	2643.4	3318.2	3363.7
Humanoid-v4	5623.9	8407.2	8277.9	7587.2	6930.8
LunarLander-v3	265.7	263.8	157.2	14.3	-77.7
Swimmer-v4	103.7	93.5	70.0	54.2	45.9
Walker2d-v4	4299.2	6306.4	5842.1	5795.2	5539.1

Learning curves In the experiments, we presented controlled study to isolate dropout’s role: while distributional pessimism eliminates overestimation bias, dropout primarily increases training stability and performance. Both learning curves are available in Appendix B. Episodic scores and related value estimation errors for varying level of pessimism under joint actor/critic dropout are summarized in Figure 5 and Figure 6. Tabular results regarding average and final performance for different pessimism levels are summarized in Table 3. In addition, same curves for different dropout configurations under same level of pessimism are summarized in Figure 7 and Figure 8, where β values are explained in Table 5. The shaded area represents quartile limits, while the solid line represents inter-quartile mean across different seeds and %1 smoothing window of evaluation episodes through environment steps.

Pessimism Table 3 summarizes final performance in terms of inter-quartile mean score on last %1 evaluation episodes. Results indicate that β is a sensitive parameter, the higher β yields a negative error (see Figure 6), consistent with our assumptions. In addition, score curves are worse if value estimations tend to be positive, meaning that critic overestimation is not mitigated enough (see Ant-v4, BipedalWalker-v3, Hopper-v4,

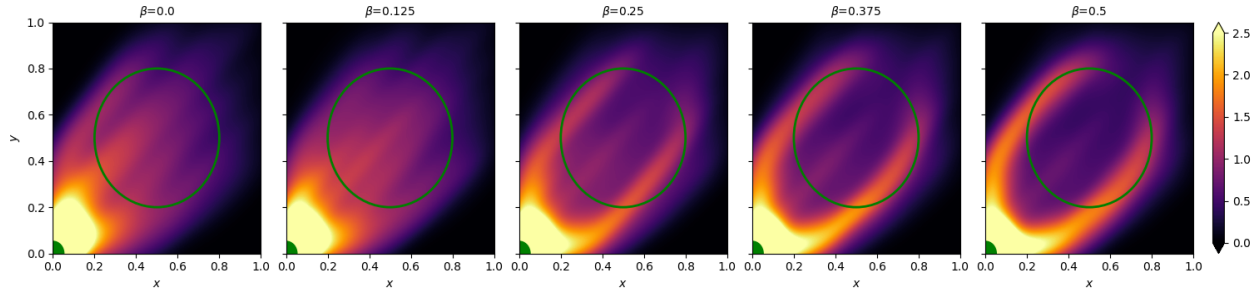


Figure 4: Position occurrence density heatmap of STAC on RiskyPointMass-v0 for different β values. The green circle on the middle represents boundary of danger zone. Episodes are terminated when point mass enters green area on the lower left.

Humanoid-v4, Walker-v4). On the other hand, score curves are again worse when error curves are negative than it should be, meaning that the agent is stuck on critic underestimation caused by high pessimism (see BipedalWalkerHardcore-v3, HalfCheetah-v4, LunarLander-v3, Swimmer-v4). In the end, pessimism sensitivity varies for different environments, possibly because of varying task difficulties. For easier tasks, lesser pessimism is enough but difficult tasks require it significantly. This parameter stands as the major bottleneck of STAC and can only be determined by this heuristic for now.

Dropout Dropout’s primary role in STAC is as a regularization tool to improve optimization stability and robustness to noisy data. In Figure 7, independent of critic dropout, actor dropout increases performance in all experiments, enhancing training stability. According to Figure 8, the overestimation trends observed with and without dropout are qualitatively similar, confirming that our core contribution—mitigating overestimation via aleatoric uncertainty—remains effective independent of dropout, except Hopper-v4. However, in this environment, learning does not even converge without critic dropout. Therefore, it is necessary in some environments for training stability rather than overestimation mitigation. Double/ensemble critic approaches mitigate training instability naturally, whereas STAC requires a convenient dropout in some cases.

5.3 Risk Sensitivity by Pessimism

In STAC, actor updates are pessimistic for one-step return uncertainty. However, critic targets are also pessimistic with respect to consecutive steps. Therefore, pessimism is also bootstrapped along with reward, and it is expected to observe increasing risk-averse behavior when higher β is used.

Following Ma et al. (2021) and Ma et al. (2025), risky navigation task is used as a simple stochastic problem. The environment is named as RiskyPointMass-v0. The goal of this task is to navigate a point mass towards a target point while avoiding a danger zone, in a two dimensional space. Danger zone is a circular area with a radius of $r_0 = 0.3$, centered at point $(0.5, 0.5)$. Initial states are randomly selected from $\mathcal{U}(0.3, 1)$ for both coordinates, excluding the danger zone. Episode terminates if the agent is close to target point $(0, 0)$ by Euclidean distance of 0.05. At each step, the agent receives sum of three reward components: negative Euclidean distance to the target, fixed -0.1 points to promote episode termination, a penalty of -10 points if the agent is within danger zone with probability $0.1 \exp(-4r^2/r_0^2)$ where r is the distance to danger zone center. The agent can move by 0.1 unit at most in both dimensions plus a random slip (at most 0.02 unit). The reward function is designed so that optimal risk-neutral solution is to navigate linearly to the target point whereas optimal risk-averse solution is to navigate around the danger zone circle.

For each β value, STAC is run on this environment with a same single seed. In Figure 4, for varying β values, position densities of the trained agents are demonstrated. The bigger green circles represent danger zone boundaries, whereas quarter green circles on the lower left represent the target points. For this demonstration, 200 evaluation steps are run with random initial points. Clearly, with higher β values, the agent prefers to go around the circle as opposed to agents trained with lower β . As expected, STAC not only mitigates overestimation but also ensures risk-averse learning with pessimism on temporal uncertainty.

6 Related Work

Overestimation reduction by epistemic uncertainty In the literature, overestimation problem is solved by pessimistic learning based on *epistemic uncertainty* of critic (Chen et al., 2021; Hiraoka et al., 2021). The same approach is also used in model-based RL methods (Janner et al., 2019; Chua et al., 2018; Depeweg et al., 2016). Recently, for off-policy model-free actor-critic algorithms, epistemic uncertainty is estimated by using double network (Fujimoto et al., 2018; Haarnoja et al., 2018) or ensemble networks (Chen et al., 2021; Moskovitz et al., 2021). Ensemble of critics are computationally expensive as there are multiple of parameters to be optimized. Using dropout is a kind of Bayesian approximation, so another way to assess epistemic uncertainty (Gal & Ghahramani, 2016). It has applications on model-based (Gal et al., 2016a; 2017; Kahn et al., 2017) and model-free (Moerland et al., 2017; Jaques et al., 2019; He et al., 2021) reinforcement learning. For actor-critic, Hiraoka et al. (2021) found out that using Bayesian dropout also reduces the necessary critic ensemble size. The mentioned methods use a constant level of pessimism for policy evaluation and improvement, while Moskovitz et al. (2021) focused on updating pessimism *on the fly* as a bandit problem rather than fixing it.

Risk-sensitive reinforcement learning Aleatoric uncertainty representation for the value function carries fundamental importance, especially in the presence of approximation (Bellemare et al., 2017). This can be conducted by atoms (Bellemare et al., 2017), quantiles (Dabney et al., 2018b; Ma et al., 2025), or a normal distribution (Tang et al., 2019; Yang et al., 2021) to model cumulative return behavior. For safety-critical RL applications to avoid catastrophic situations, aleatoric pessimism is used by Tang et al. (2019) and Yang et al. (2021). Different than STAC, these methods use pessimism only on actor update for safety, not for overestimation reduction, and models full return uncertainty.

Overestimation reduction by aleatoric uncertainty Kuznetsov et al. (2020) claims *aleatoric uncertainty* is also responsible for overestimation since any randomness is exploited when the Bellman optimality operator ($\mathcal{T}^*$) is employed. For this purpose, they use ensemble networks for epistemic uncertainty, in which each network is a distributional network (as quantiles) for aleatoric uncertainty assessment, and used both types of uncertainties for overestimation correction. Similarly, STAC uses distributional critic, but only models temporal (one-step) aleatoric uncertainty rather than full return distribution as this is the main source of overestimation. It is also not epistemic uncertainty since it is not related about weights of the critic network.

7 Conclusion & Future Directions

In this paper, we introduced Stochastic Actor-Critic (STAC), a novel off-policy actor-critic algorithm. The main idea is to mitigate overestimation for the sake of faster and more robust learning by incorporating the pessimistic learning objective using temporal aleatoric critic uncertainty. For this, critic is modeled as a distributional neural network unlike previous approaches. Although normal distribution is used for this purpose, our analysis is valid for all sub-Gaussian critic representations.

We derived an upper bound for overestimation, demonstrating that an adequate level of pessimism mitigates overestimation without succumbing to underestimation, thus facilitating computation and sample-efficient learning. As a by-product, this approach also yields risk-averse learning process. Lastly, using dropout on actor and critic networks are proposed to mitigate over-fitting. Actor dropout always improves performance, whereas using dropout on critic is only required to ensure learning convergence.

Adaptive Pessimism Our ablation studies demonstrate the effects of dropout rate and pessimism, revealing the sensitivity of the learning procedure to these parameters. For each specific environment and optimization method, an optimal level of pessimism and dropout exists. A promising direction for future research is to develop a grounded method to adjust the pessimism level for specific environments and agents to allow better adaptation for the learner to the environment, similar to Moskovitz et al. (2021); Cetin & Celiktutan (2023).

Distribution Modeling For better modeling, aleatoric critic uncertainty can be modeled by quantiles (Dabney et al., 2018b; Ma et al., 2025) or flow matching (Chen et al., 2025; Dong et al., 2025), rather than normal distribution.

Pessimism under Stochastic Environments The effect of pessimism on highly stochastic environments is also an important topic for research. As shown, pessimistic updates lead to risk-averse behavior. Our results are limited to simple stochastic tasks, and it is worth to investigate STAC empirically on more advanced ones.

Exploration Current methods use epistemic critic uncertainty for overestimation problem, which contradicts with *optimism in the face of uncertainty* principle. Future researches might consider using epistemic critic and actor uncertainty for exploration, retaining temporal aleatoric pessimism.

Broader Impact STAC is an important contribution to understand overestimation phenomenon in off-policy actor-critic learning. It demonstrates that critic overestimation is a topic of distributional learning, and the solution is to devise a risk-averse learning objective for both actor and critic. Moreover, it also proves that modeling epistemic uncertainty is not necessary at all for this problem, improving computation efficiency directly.

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Appendix A Proofs

Proof of Theorem 3.1. Analyzing Bellman update $\mathcal{T}^*Q(s, a)$,

$$\begin{aligned}
\mathcal{T}^*Q(s, a) &= R(s, a) + \gamma \mathbb{E}_{Q \sim \mathcal{Q}} \left[\tilde{\alpha} \log \left(\int_{\mathcal{A}} \exp(\tilde{\alpha}^{-1} Q(s', a')) da' \right) \right] \\
&\leq R(s, a) + \gamma \tilde{\alpha} \log \left(\mathbb{E}_{Q \sim \mathcal{Q}} \left[\int_{\mathcal{A}} \exp(\tilde{\alpha}^{-1} Q(s', a')) da' \right] \right) \\
&= R(s, a) + \gamma \tilde{\alpha} \log \left(\int_{\mathcal{A}} \mathbb{E}_{Q \sim \mathcal{Q}} \left[\exp(\tilde{\alpha}^{-1} Q(s', a')) \right] da' \right) \\
&\leq R(s, a) + \gamma \tilde{\alpha} \log \left(\int_{\mathcal{A}} \exp(\tilde{\alpha}^{-1} \mu(s', a') + \frac{1}{2} \tilde{\alpha}^{-2} \sigma^2(s', a')) da' \right)
\end{aligned}$$

First inequality is by Jensen's inequality (using concave property of log function) while the following equality is a result of Tonelli's theorem. The second inequality results from the sub-Gaussian assumption 3.1. $\square$

Proof of Corollary 3.1.1. From the Theorem 3.1, we can show that

$$\begin{aligned}
\mathcal{T}_\beta^*Q(s, a) &= R(s, a) + \gamma \mathbb{E}_{Q \sim \mathcal{Q}} \left[\tilde{\alpha} \log \left(\int_{\mathcal{A}} \exp(\tilde{\alpha}^{-1} Q(s', a') - \beta \sigma(s', a')) da' \right) \right] \\
&\leq R(s, a) + \gamma \tilde{\alpha} \log \left(\int_{\mathcal{A}} \exp(\tilde{\alpha}^{-1} (\mu(s', a') - \beta \sigma(s', a') + \frac{1}{2} \tilde{\alpha}^{-1} \sigma^2(s', a'))) da' \right) \\
&= R(s, a) + \gamma \tilde{\alpha} \log \left(\int_{\mathcal{A}} \exp(\tilde{\alpha}^{-1} \mu^\dagger(s', a')) da' \right) = \mathcal{T}^* \mu^\dagger(s, a).
\end{aligned}$$

where we have defined $\mu^\dagger(s', a') = \mu(s', a') - \beta \sigma(s', a') + \frac{1}{2} \tilde{\alpha}^{-1} \sigma^2(s', a')$. If $\beta \geq \max_{(s', a')} \frac{1}{2} \tilde{\alpha}^{-1} \sigma(s', a')$, then $\mu^\dagger(s', a') < \mu(s', a')$. So we can show that

$$\mathcal{T}_\beta^*Q(s, a) \leq \mathcal{T}^* \mu^\dagger(s, a) \leq \mathcal{T}^* \mu(s, a). \quad (16)$$

$\square$

Proof of Theorem 3.2. From the Theorem 3.1, we can show that

$$\begin{aligned}
\mathbb{E}_{s' \sim \tau} [\mathcal{T}^*Q(s, a)] &\leq R(s, a) + \gamma \mathbb{E}_{s' \sim \tau} \left[\tilde{\alpha} \log \left(\int_{\mathcal{A}} \exp(\tilde{\alpha}^{-1} \mu(s', a') + \frac{1}{2} \tilde{\alpha}^{-2} \sigma^2(s', a')) da' \right) \right] \\
&\leq R(s, a) + \gamma \mathbb{E}_{s' \sim \tau} \left[\tilde{\alpha} \log \left(\left(\int_{\mathcal{A}} \exp(\tilde{\alpha}^{-1} \mu(s', a')) da' \right) \cdot \left(\max_{a'} \exp(\frac{1}{2} \tilde{\alpha}^{-2} \sigma^2(s', a')) \right) \right) \right] \\
&= R(s, a) + \gamma \mathbb{E}_{s' \sim \tau} \left[\tilde{\alpha} \log \left(\int_{\mathcal{A}} \exp(\tilde{\alpha}^{-1} \mu(s', a')) da' \right) \right] + \frac{\gamma}{2\tilde{\alpha}} \mathbb{E}_{s' \sim \tau} \left[\max_{a'} \sigma^2(s', a') \right].
\end{aligned}$$

The second inequality is a result of the mean value theorem for integrals. In the last equality, the first two terms are equal to $\mathcal{T}^* \mu(s, a)$. Therefore,

$$\epsilon(s, a) = \mathbb{E}_{s' \sim \tau} [\mathcal{T}^*Q(s, a)] - \mathcal{T}^* \mu(s, a) \leq \frac{\gamma}{2\tilde{\alpha}} \mathbb{E}_{s' \sim \tau} \left[\max_{a'} \sigma^2(s', a') \right].$$

$\square$

Appendix B Results of Ablation Studies

B.1 Pessimism Ablation

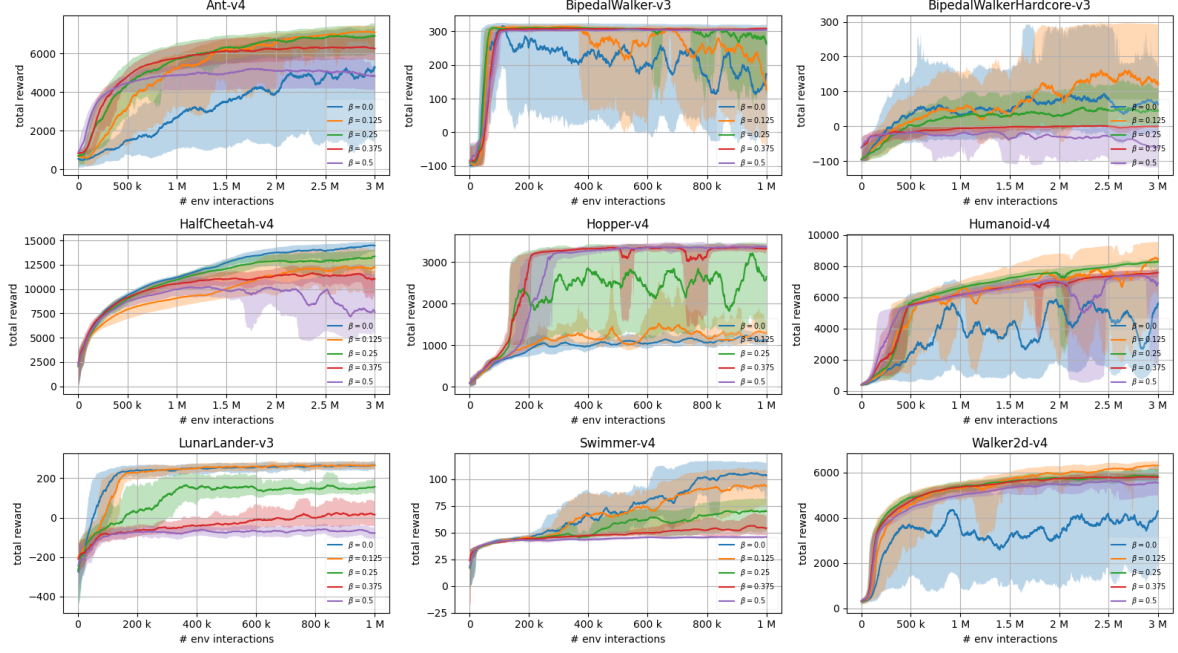


Figure 5: Episodic score curves of STAC with varying pessimism (β) parameter, with actor and critic dropout equal to 0.01.

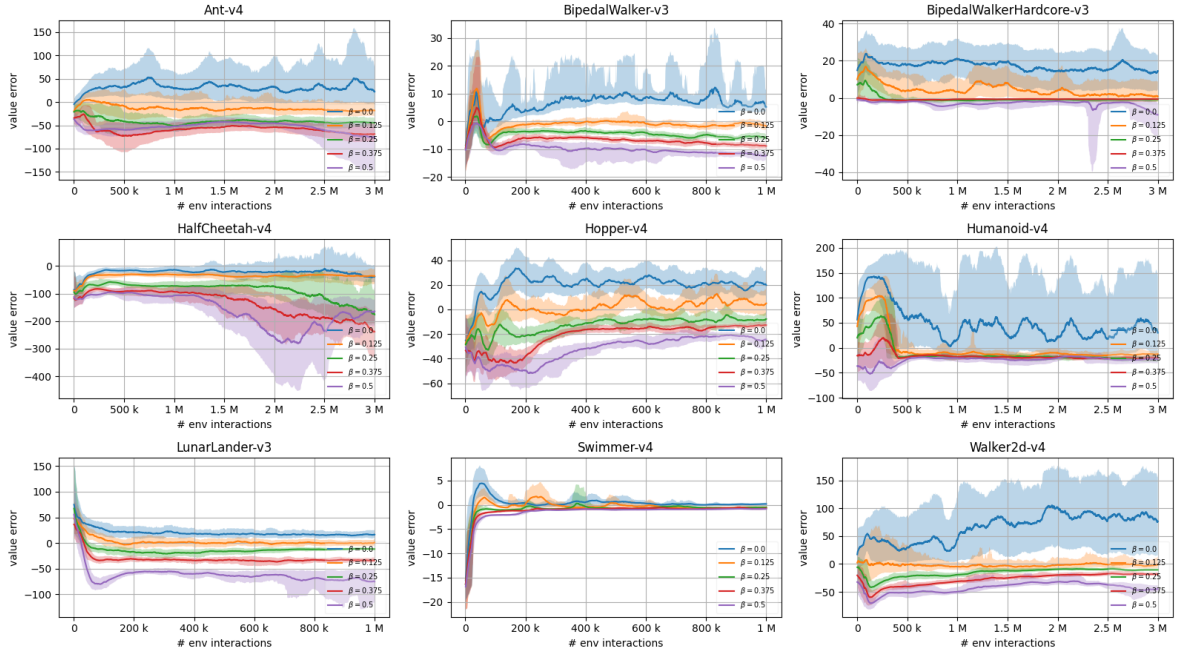


Figure 6: Average episodic value estimation error curves of STAC with varying pessimism (β) parameter, with actor and critic dropout equal to 0.01.

B.2 Dropout Configuration Ablation

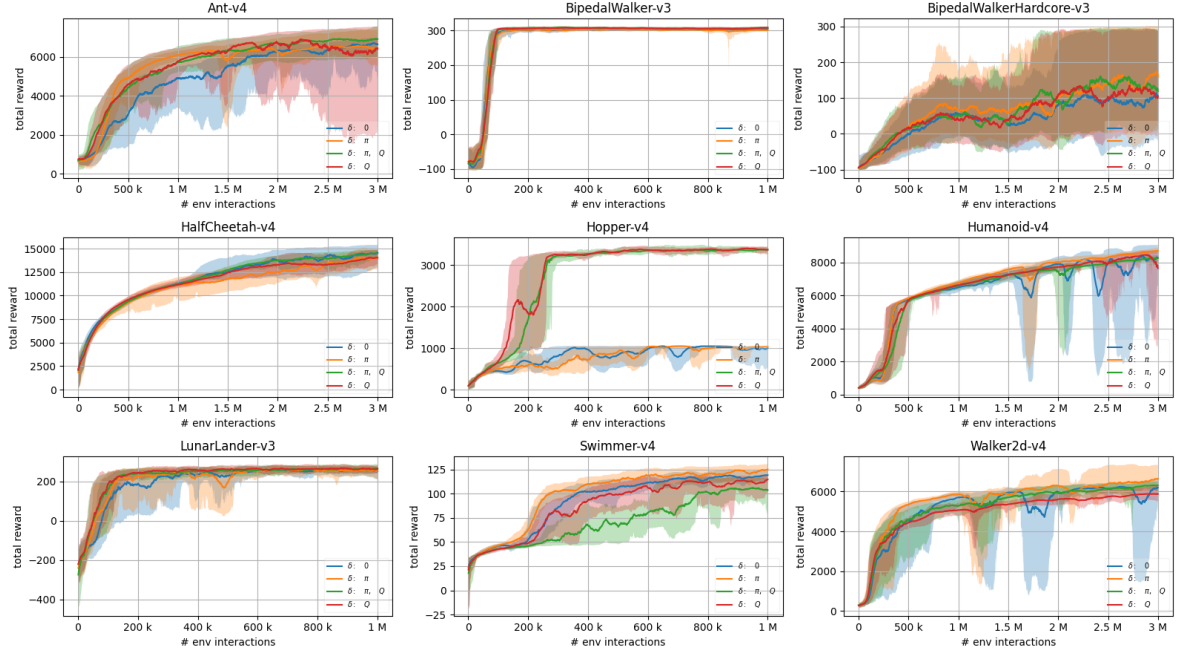


Figure 7: Episodic score curves of STAC with 4 different dropout configurations, under fixed pessimism level yielding best performance (see Table 5).

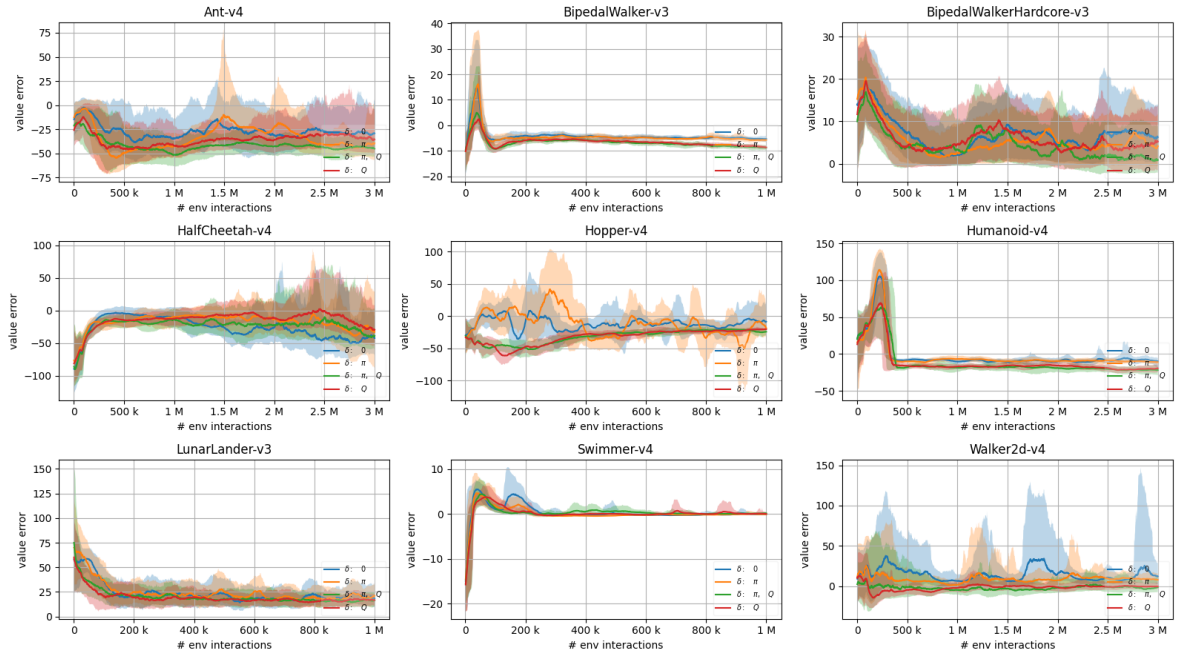


Figure 8: Average episodic value estimation error curves of STAC with 4 different dropout configurations, under fixed pessimism level yielding best performance (see Table 5).

Appendix C Hyper-parameters and Experiment Details

Table 4: Experimental Parameters per Algorithm

Algorithm	Parameter	Value
DSAC, ESTAC, SAC, STAC	Optimizer	Adam ((Kingma & Ba, 2014))
	Critic Learning Rate	3×10^{-4}
	Actor Learning Rate	3×10^{-4}
	Discount Rate (γ)	0.99
	Target-Smoothing Coefficient (ρ)	0.995
	Target Entropy ($\bar{H}$)	$- \mathcal{A} $
	Replay Buffer Size	1×10^6
	Mini-Batch Size	256
	Learning Starting Step	10000
	UTD Ratio (G)	1
DSAC	Number of Quantiles	25
	Huber Loss threshold (κ)	1.0

Table 5: Target policy entropy ($\bar{H}$), pessimism β and dropout rates per environment, yielding best results. All algorithms share same policy entropy.

Environment	Entropy ($\bar{H}$)	Pessimism (β)	Actor Dropout	Critic Dropout
Ant-v4	-4	0.25	0.01	0.01
BipedalWalker-v3	-2	0.375	0.01	0.01
BipedalWalkerHardcore-v3	-2	0.125	0.01	0
HalfCheetah-v4	-3	0.0	0.01	0.01
Hopper-v4	-1	0.5	0.01	0.01
Humanoid-v4	-8	0.25	0.01	0
LunarLander-v3	-2	0.0	0.01	0.01
Swimmer-v4	-1	0.0	0.01	0
Walker2d-v4	-3	0.125	0.01	0

Appendix D Network Architectures of STAC

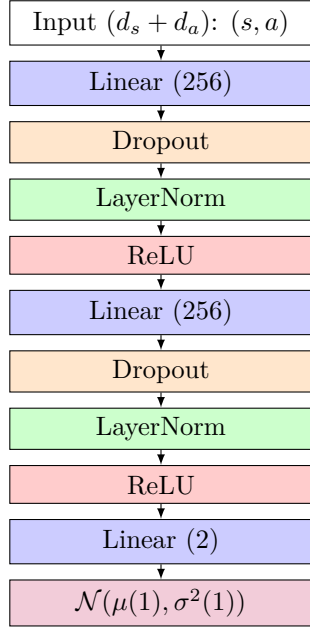


Figure 9: Critic network architecture

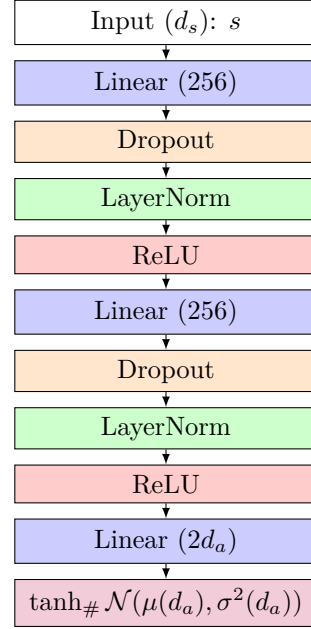


Figure 10: Actor network architecture

Appendix E Source Code

Our results can be accessed publicly at <https://github.com/authors-github/stochastic-actor-critic-results>. This code uses our in-house developed RL framework as a sub-repository, available on <https://github.com/authors-github/rl-warehouse>.